THE UNIVERSITY OF DANANG
THE UNIVERSITY OF DANANG – UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF ELECTRONICS AND ARTIFICIAL INTELLIGENCE







TECHNICAL REPORT

DESIGN OF A MULTI-MODE PWM MONITORING AND CONTROL SYSTEM VIA UART INTERFACE

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Da Nang, August 2026

ABSTRACT

As part of the FPGA MCU Design Danang 2026 competition, I carried out the practical implementation using an FPGA development board provided by the company. Among the two available hardware platforms, namely the Kiwi 1P5, featuring the Gowin GW1N-UV1P5 FPGA, and the Kiwi Nano 4K, featuring the GW1NSR-LV4C FPGA, I selected the Kiwi 1P5 board as the platform for this project. This selection provided the basis for studying the board schematic, identifying the required I/O pins, and configuring the hardware according to the system design.

Using the Kiwi 1P5 platform, I designed and implemented a multi-mode PWM monitoring and control system via UART communication using Verilog HDL, GOWIN EDA, and the Hercules Serial Terminal software. The system is designed based on a synchronous digital architecture and consists of several functional blocks, including a PLL-based system clock generator, debounce circuits for two push buttons, an LED PWM controller, a Finite State Machine (FSM), and a UART transmitter. The LED operates in three modes: LOW with a 25% duty cycle, HIGH with a 100% duty cycle, and AUTO with a breathing LED effect completed within 2 seconds.

In addition to LED control, the system uses a UART interface configured at 115200 bps, 8 data bits, no parity, and 1 stop bit to transmit system status information from the FPGA to a computer. Whenever the system starts up or changes its operating mode, the FPGA transmits the corresponding messages, such as “MODE: LOW”, “MODE: HIGH”, and “MODE: AUTO”, through the UART interface. Therefore, this practical implementation not only focuses on FPGA programming but also provides an opportunity to integrate and evaluate push-button processing, PWM control, FSM design, and UART peripheral communication on actual FPGA hardware.

This report presents the process of selecting and studying the Kiwi 1P5 development board, designing the system architecture and functional modules, developing the FSM state-transition diagram, calculating the UART baud-rate divider, and performing simulation and practical implementation on the FPGA. The results provide a basis for evaluating the completed practical work and preparing for the FPGA MCU Design Danang 2026 competition.

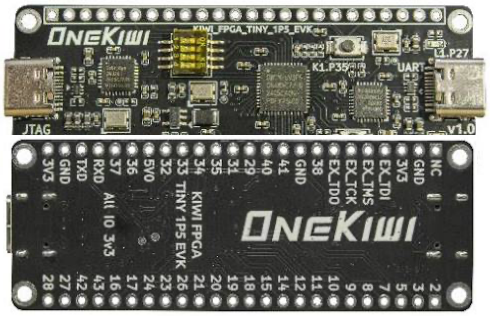
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1. KIWI 1P5 CIRCUIT BOARD



- Main FPGA Chip: GW1N-UV1P5 on chip Flash
- Embedded 1584 FPGA LUTs
- Onboard Gowin GWU2X USB-JTAG debugger
- Onboard Button and LED
- On-chip Flash – no need external Flash IC
- Support JTAG DIP Switch program external FPGA

Figure 1.1: Kiwi 1P5 board.

		Kiwi 1P5
FPGA Details	**FPGA Chip**	GW1N-UV1P5 **GW1N-UV1P5QN48XC7/I6**
	Logic Units	1584
	Hard processor	None
	Block SRAM (bits)	96K
	User Flash (bits)	256K
	Number of PLL	2
	Number of I/O Bank	6
	Number of user I/O	125
User Interface	**User Button**	02
	User LED	02
Programmer	**Programmer IC**	Gowin U2X Support Programming for External FPGA (Change Switch SW1)
Physical	**Size**	70mm x 22 mm
	Interface	Double row header pitch 2.54mm
Power	**Main Voltage**	5V
	I/O Voltage	All IO 3.3V
Environment	**Temperature**	Industrial: - 40°C to 75°C
	Humidity	Humidity (non-condensing):10% - 90%

Table 1.1: General specifications of the Kiwi 1P5 circuit board.

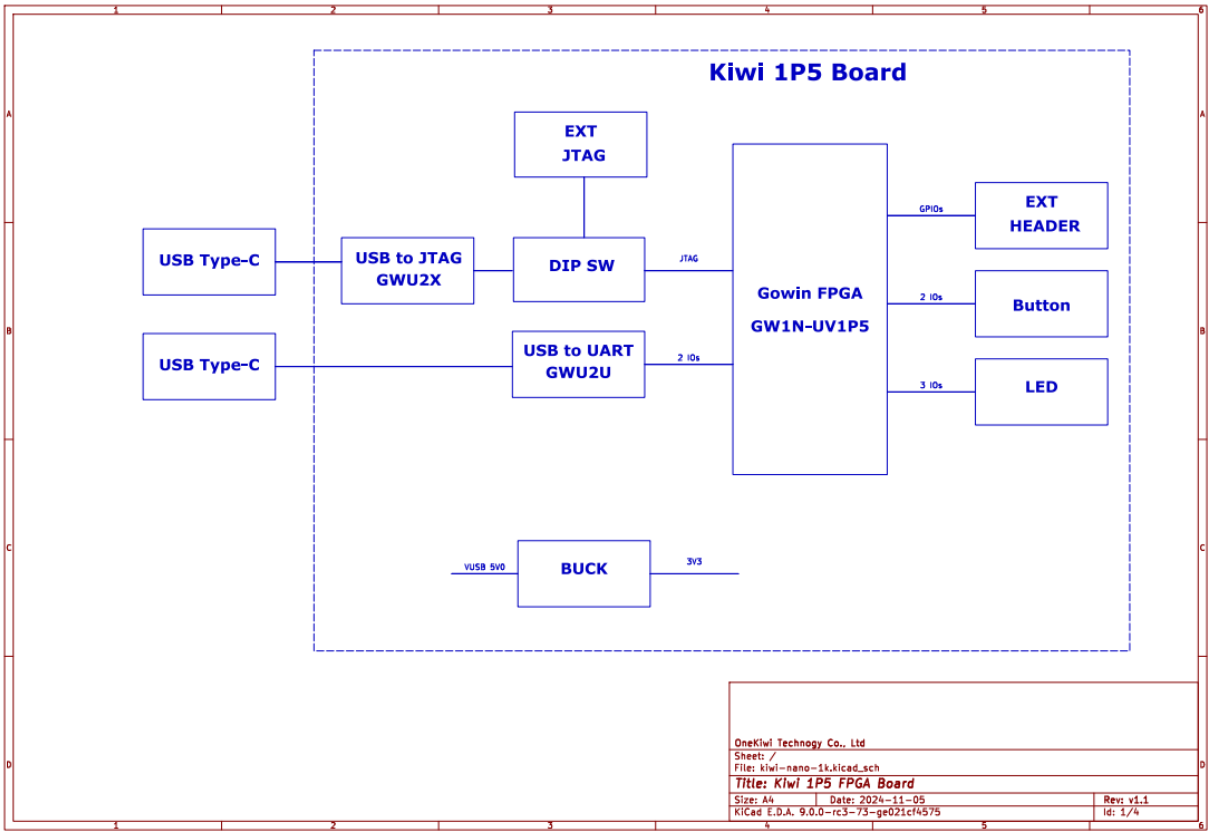


Figure 1.2: Kiwi 1P5 hardware block diagram: GW1N-UV1P5 FPGA, GWU2X USB-JTAG, GWU2U USB-UART, push-buttons, and LEDs.

The table below summarizes the pins used directly in the source code and .cst file.

Signal	FPGA Pin	Bank	I/O Type	Function
clk_27m	4	5	Input, LVCMOS33, Pull-up	Clock onboard 27 MHz
btn1	35	1	Input, LVCMOS33, Pull-up	Button 1, active-low
btn2	36	1	Input, LVCMOS33, Pull-up	Button 2, active-low
led	27	1	Output, LVCMOS33, Drive 8	PWM-controlled LED
uart_tx	34	1	Output, LVCMOS33, Drive 8	UART TX → GWU2U

Table 1.2: Pin Assignments and I/O Configurations

According to the pinout/schematic file, pin 35 corresponds to IOR1B, pin 36 to IOR1A, pin 27 to IOR17A, and pin 34 to IOR11A. The I/O voltage level is 3.3 V. The board's two push-buttons are active-low by default, which aligns with the debounce logic in the code.

2. SYSTEM ARCHITECTURE

The system architecture is strictly based on Synchronous Digital Design principles, where all internal modules operate within a shared clock domain generated by a central clock management block. The design features a top-level module (top_system) that routes data and interconnects all functional sub-modules.

2.1. System Block Diagram

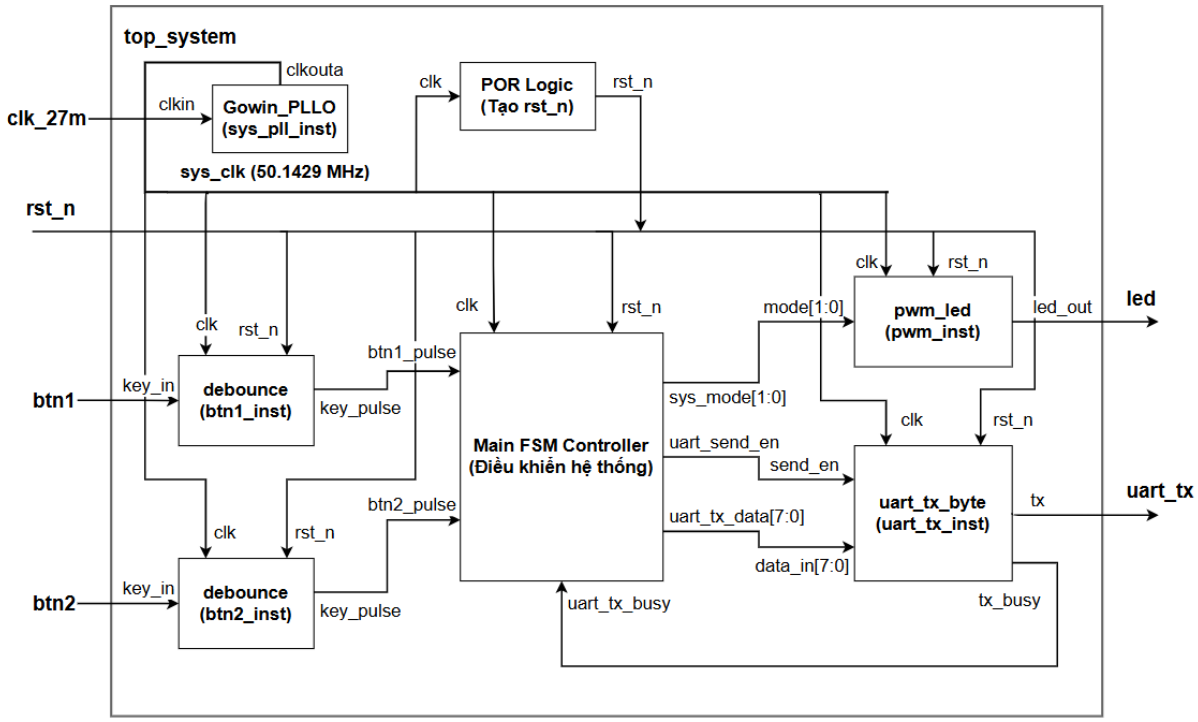


Figure 2.1: System block diagram of the top_system.

According to the top_system block diagram, the architecture consists of the following primary functional blocks:

- **Gowin_PLLO Block (sys_pll_inst):** Takes the 27 MHz clock input (clk_27m) from the onboard oscillator to synthesize a stable 50.1429 MHz internal system clock (sys_clk).
- **POR Logic Block (Power-On-Reset):** Generates an active-low internal reset signal (rst_n) during the first 1.3ms after power-up. This circuit ensures that the entire system (especially the FSM registers) begins operation only after the power supply and PLL clock have completely stabilized.
- **Debounce Block (btn1_inst & btn2_inst):** Filters out mechanical bouncing noise from the two input buttons (btn1, btn2). This block outputs single-cycle

active pulses (btn1_pulse, btn2_pulse) upon detecting a valid button press, preventing the subsequent FSM from capturing erroneous states.

- **Main FSM Controller Block:** Acts as the brain of the system, receiving button pulses to determine the current operating mode (sys_mode[1:0]). This block also handles the UART string transmission state machine, automatically loading data (uart_tx_data[7:0]) and triggering the transmission enable signal (uart_send_en) based on the busy flag (tx_busy) fed back from the UART block.
- **PWM_LED Block (pwm_inst):** Receives the mode control code (sys_mode[1:0]) from the Main FSM to compute the corresponding Duty cycle. It then outputs a logic level signal directly to the hardware pin to drive the LED brightness (led_out).
- **UART_TX_BYTE Block (uart_tx_inst):** The asynchronous serial transmission block. It takes a data byte and an enable pulse from the FSM, performs parallel-to-serial conversion, and transmits the bitstream through the uart_tx pin to communicate with the host PC.

2.2. Data and Control Flow

The operation of the entire system is divided into two independent yet tightly synchronized paths, coordinated by the central state machine:

Control Flow: The control signals originate from physical interactions with the two push-buttons. These inputs (btn1, btn2) are routed through the Debounce block to eliminate mechanical noise, generating clean single-cycle pulses (btn1_pulse, btn2_pulse). The Main FSM uses these pulses as transition conditions to latch and output the system's global operating mode (sys_mode[1:0]).

Data Flow: This path branches into two directions:

- **PWM LED Path:** The PWM block receives the sys_mode code to select the corresponding Duty Cycle threshold (25%, 100%, or auto-increment/decrement), directly outputting the hardware control signal led_out.
- **UART TX Path:** Whenever the sys_mode changes, a secondary state machine within the Main FSM triggers the string transmission process. The FSM sequentially loads each ASCII character (uart_tx_data[7:0]) and asserts the send_en pulse to the UART_TX_BYTE block. The UART block receives the byte, performs parallel-to-serial conversion, and shifts it out via the uart_tx pin. This process is perfectly synchronized using the tx_busy handshaking flag, which feeds back to the FSM to indicate when the next byte can be

transmitted.

3. DETAILED MODULE DESIGN

3.1. Clock Management Block

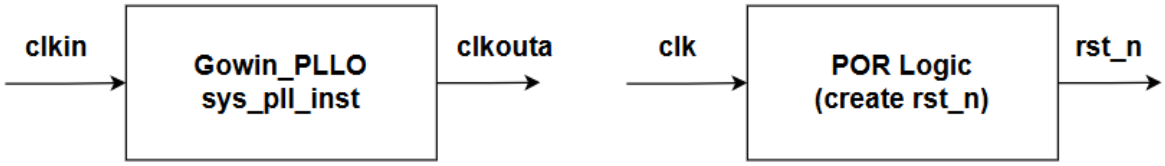


Figure 3.1: Block diagram of the Clock Management module

PLL System Clock: Utilizes the Gowin PLLO IP Core. The input frequency from the onboard oscillator is 27 MHz. By configuring the VCO divider to 7 and the feedback multiplier (CLKFB) to 13, the module generates an actual output frequency of 50.1429 MHz with a tolerance of 0.5%.

Power-On-Reset (POR) Circuit: Implements a 16-bit counter to count up from 0 to 0xFFFF. The internal reset signal (rst_n) is held low for the first ~1.3ms, ensuring the entire system only begins operation after the PLL clock is completely stable.

3.2. Debounce Block

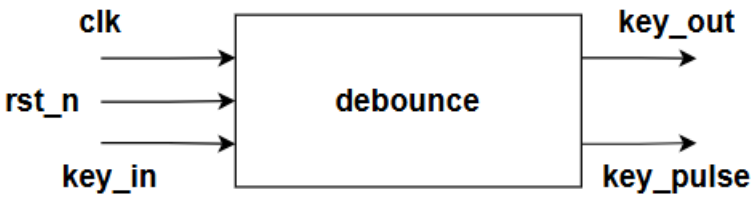


Figure 3.2: Block diagram of the Debounce module.

Synchronization: The raw button input (key_in) is passed through a 2-stage shift register (key_sync_0, key_sync_1) clocked by the system clock to eliminate metastability issues.

Noise Filtering: Uses a count register with a delay threshold of DELAY_CYCLES = 1,002,857 (exactly 20ms at 50.1429 MHz). The output is updated only if the button state remains perfectly stable for this 20ms duration.

Edge Detection: A falling-edge detection logic (key_out_reg == 1 and key_out == 0) generates a single-cycle active pulse (key_pulse), which strictly triggers FSM transitions without looping.

3.3. PWM LED Controller Block

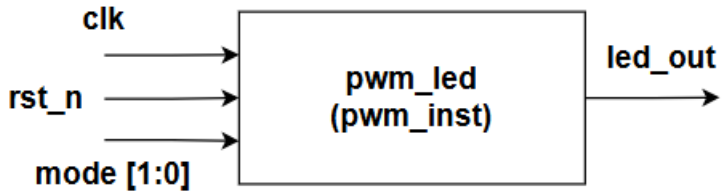


Figure 3.3: Block diagram of the PWM LED Controller module.

PWM Frequency: The pwm_cnt counts from 0 to PWM_MAX - 1 (where PWM_MAX = 50,143), establishing a base PWM frequency of 1 kHz (1ms period) to completely prevent human-eye flickering.

Multi-mode Control:

- **LOW (00):** Fixed duty cycle at 25% (PWM_MAX / 4).
- **HIGH (01):** Fixed duty cycle at 100% (PWM_MAX).
- **AUTO (10):** Breathing effect driven by the breath_cnt. The duty cycle threshold is incremented or decremented by STEP_VAL = PWM_MAX / 1000 every 1ms. It takes 1 second to fade in (0% to 100%) and 1 second to fade out (100% to 0%), resulting in a precise 2.0-second cycle.

3.4. UART TX Block (UART Transmitter)

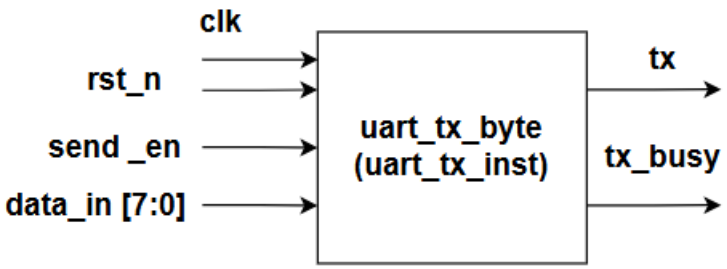


Figure 3.4: Block diagram of the UART TX module.

Configuration: Asynchronous serial transmission operating at 115200 bps, 8 Data bits, 1 Stop bit, No Parity.

Transmission State Machine: The process involves 4 core states (IDLE, START, DATA, STOP), asserting a tx_busy flag to prevent data collision during active transmission.

Baudrate Calculation: The frequency divider is derived from the PLL system clock.

Parameter	Calculation / Value	Notes
System Clock (f_{clk})	50,142,900 Hz	Output from the PLL
Target Baudrate (B_{target})	115,200 bps	Required specification
Theoretical Divider (N)	$N = \frac{50142900}{115200} \approx 435.268$	Frequency division formula
Configured Divider (BAUD_DIV)	**435**	Integer value for the counter
Actual Baudrate (B_{actual})	$B_{actual} = \frac{50142900}{435} \approx 115{,}271$ bps	Hardware output speed
Tolerance (Error)	$Error = \frac{115271 - 115200}{115200} \times 100\% \approx +0.06\%$	Below the < 3% threshold

Table 3.1: Baud Rate Divider Calculation based on 50.1429 MHz System Clock.

3.5. Main FSM Controller Block

System Monitor FSM

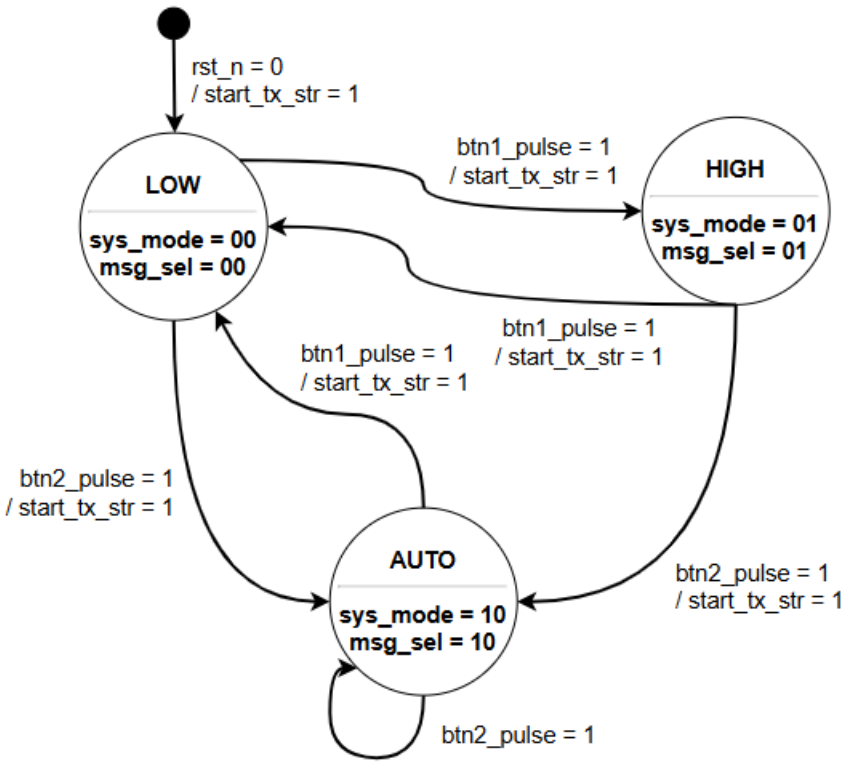


Figure 3.5: State transition diagram of the System Mode FSM.

Operation: The system initializes in the LOW state upon power-up.

Transitions: The btn1_pulse toggles the system between LOW and HIGH, and returns it to LOW if currently in AUTO. The btn2_pulse strictly forces an immediate transition to AUTO.

Trigger: Any valid state transition automatically asserts the start_tx_str flag to initiate UART reporting.

UART String FSM (Auto-transmitter)

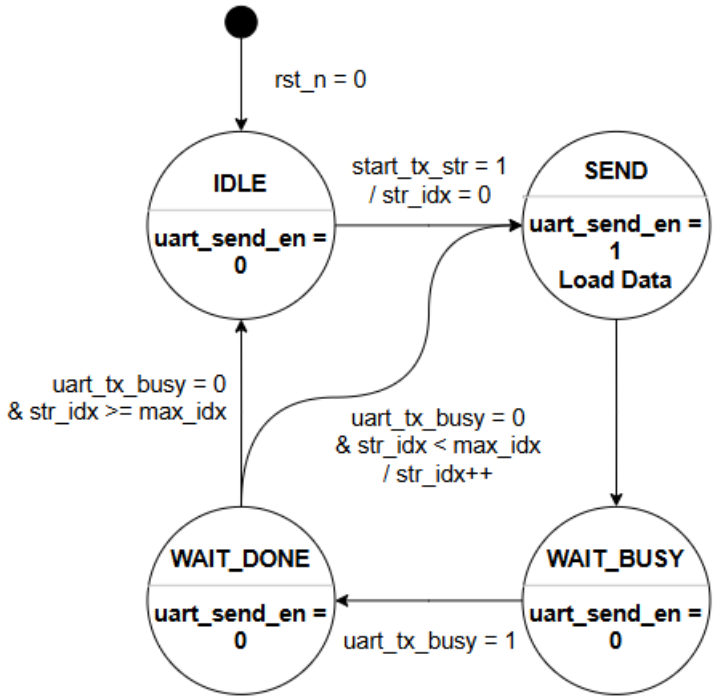


Figure 3.6: State transition diagram of the UART String Auto-transmitter.

Operation: Comprises 4 states (IDLE, SEND, WAIT_BUSY, WAIT_DONE) to handle string transmission sequentially.

Handshaking: It maps the current index (str_idx) to the corresponding ASCII character (e.g., "MODE: LOW\r\n") and uses the tx_busy signal from the UART_TX module to coordinate byte-by-byte transmission.

Dynamic Constraint: The transmission string length adapts automatically based on the active mode (10 characters for LOW, 11 characters for HIGH/AUTO).

4. SIMULATION RESULTS

4.1. Debounce Module Simulation

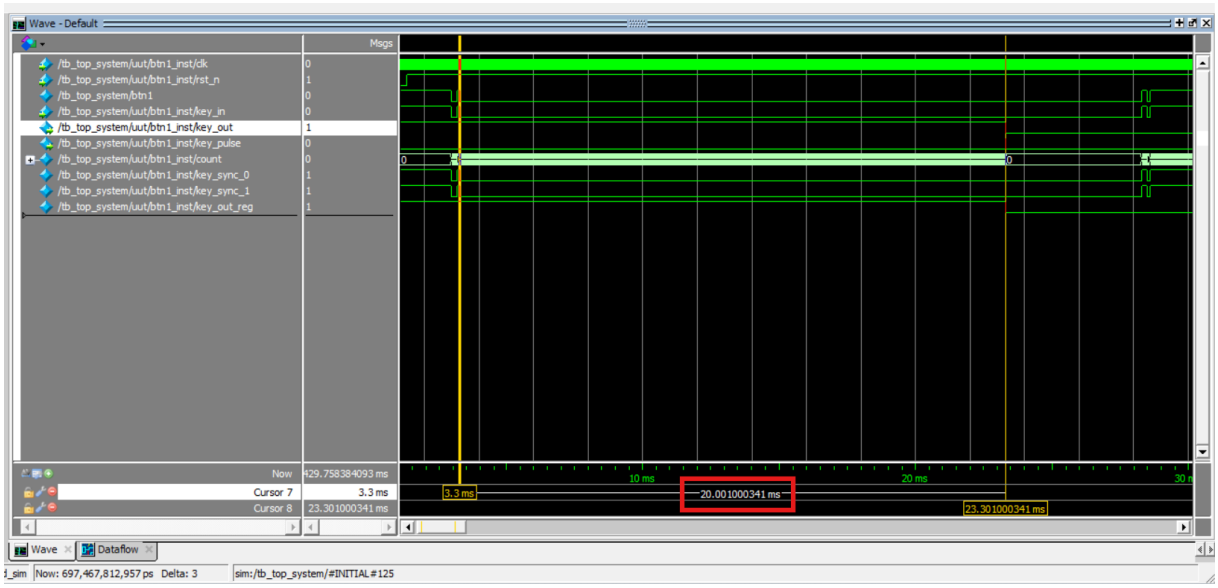
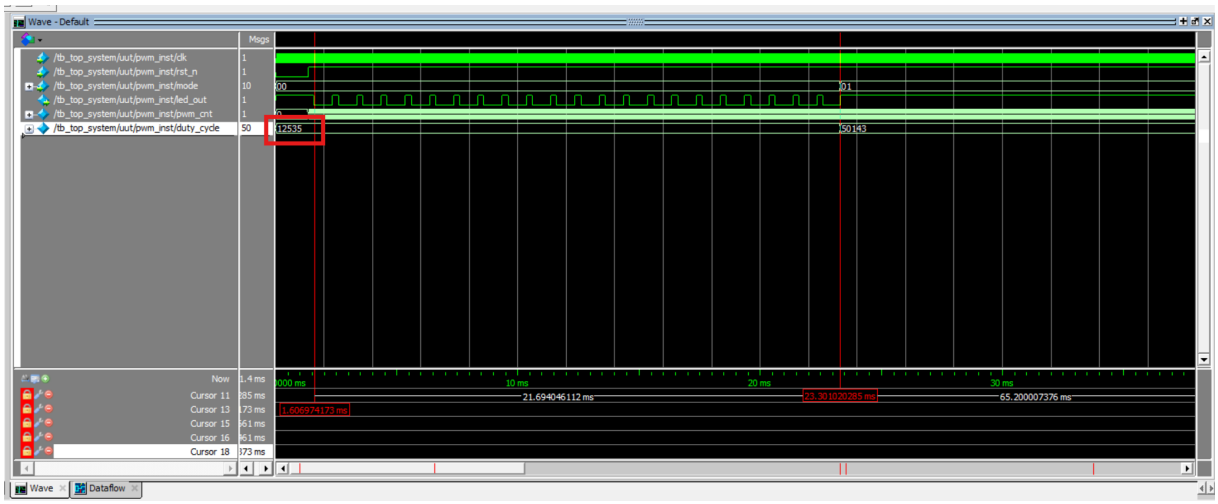


Figure 4.1: Simulation waveform of the Debounce block

Result: The simulation confirms that the module effectively filters out high-frequency mechanical bouncing (key_in). The active key_pulse is accurately asserted for exactly one clock cycle only after the input signal remains stable for the defined 20ms threshold.

4.2. PWM LED Simulation



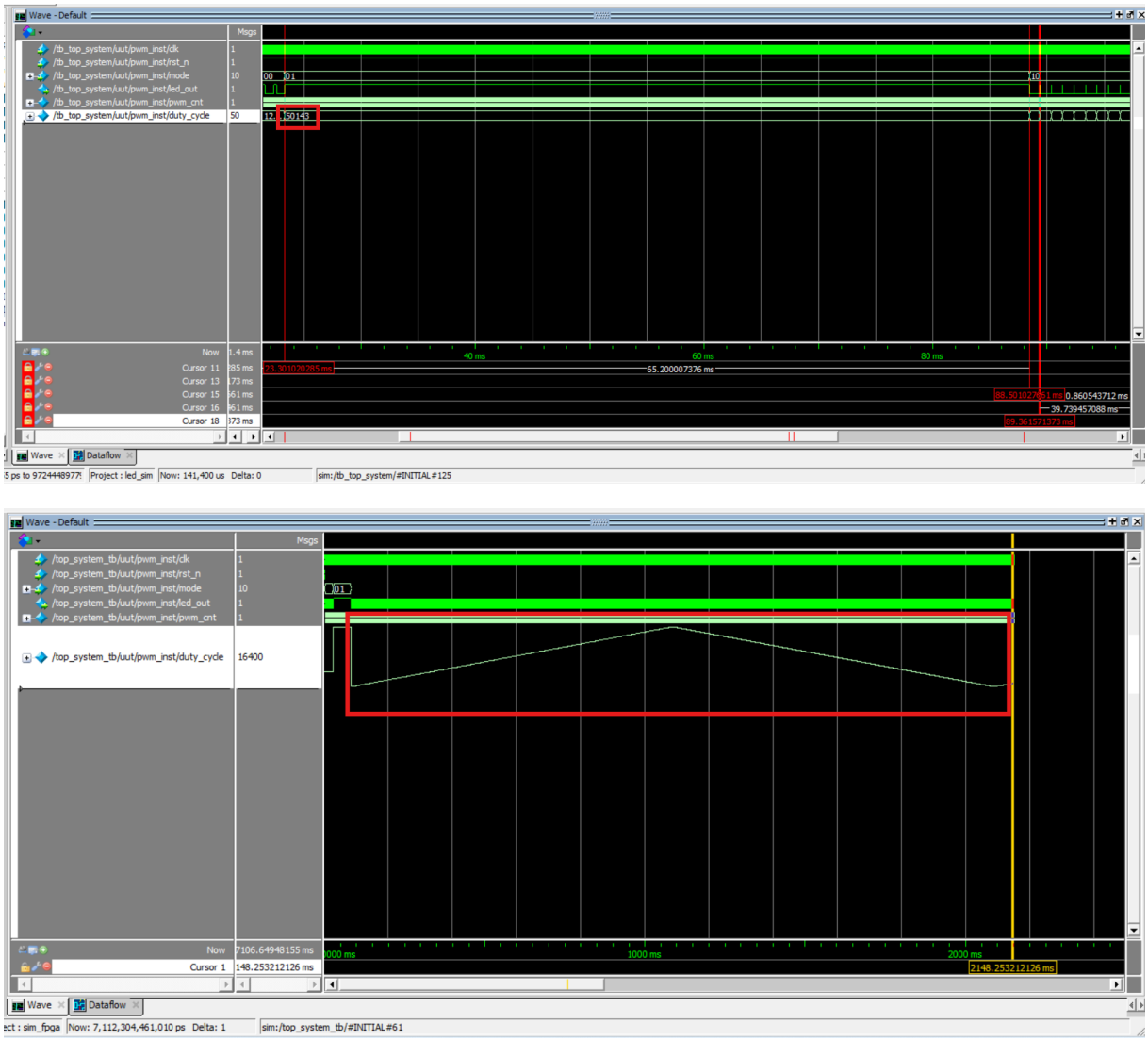


Figure 4.2: Simulation waveform of the PWM duty cycles

Result: The waveform demonstrates precise duty cycle adjustments according to the system mode. It shows a stable 25% duty cycle for the LOW mode, 100% for the HIGH mode, and a seamless, auto-incrementing/decrementing duty cycle over a 2.0-second period for the AUTO (breathing) mode.

4.3. UART TX String Simulation

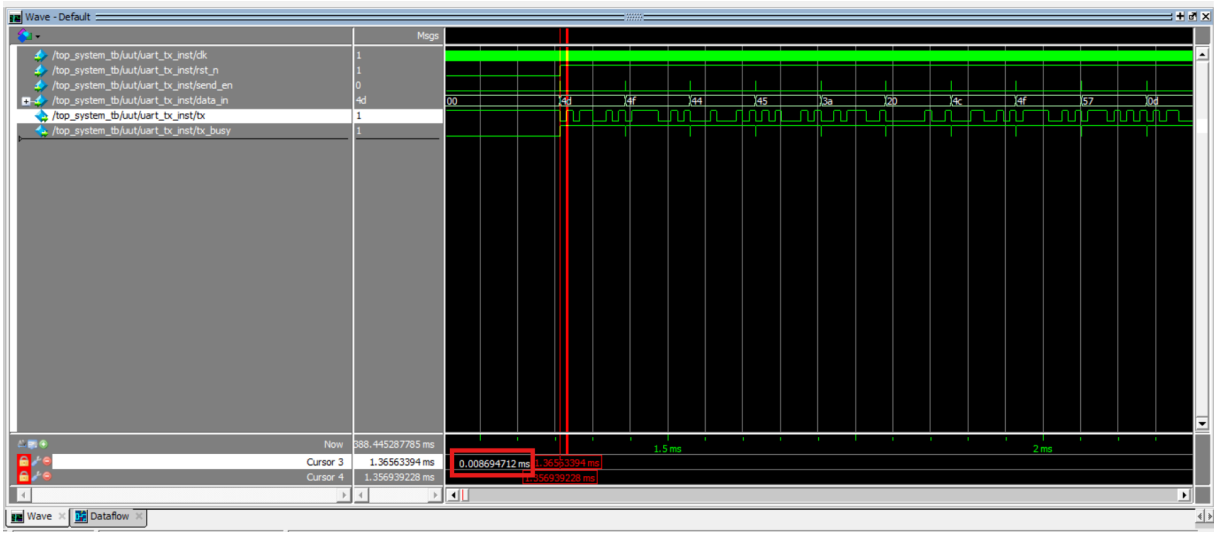


Figure 4.3: Simulation waveform of the UART string transmission.

Result: The simulation verifies that the Main FSM accurately coordinates with the tx_busy flag to transmit the correct ASCII sequences (e.g., "MODE: LOW\r\n"). The signal timing aligns perfectly with the 115200 bps baud rate, strictly adhering to the 8N1 frame format (1 Start bit, 8 Data bits, 1 Stop bit).

5. HARDWARE IMPLEMENTATION

5.1. Physical Constraints and Synthesis

The design is successfully synthesized and mapped to the KIWI 1P5 development board (GW1N-UV1P5). The I/O assignments strictly follow the physical constraints: clk_27m (Pin 4), btn1 (Pin 35), btn2 (Pin 36), led (Pin 27), and uart_tx (Pin 34).

5.2. On-Board Operation

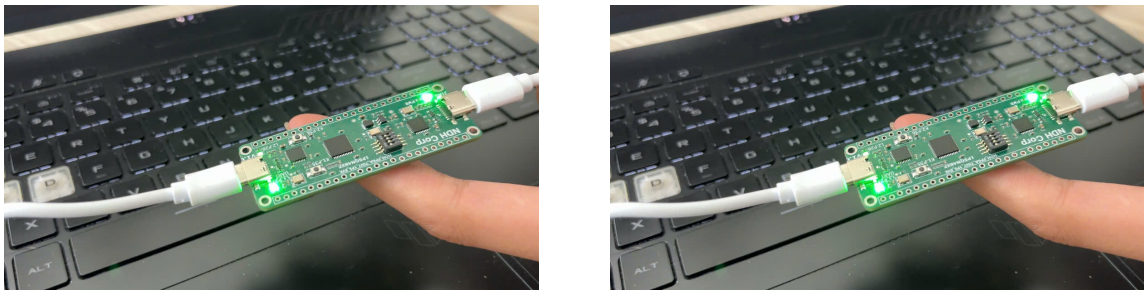


Figure 5.2: Hardware setup and LED mode demonstration.

Result: The system operates flawlessly on the hardware. The LED visually reflects the LOW (dim) and HIGH (bright) states accurately, while the AUTO mode displays a smooth 2.0-second breathing effect without any visible flickering. The

hardware debounce logic ensures clean mode transitions upon button presses.

5.3. UART Serial Communication

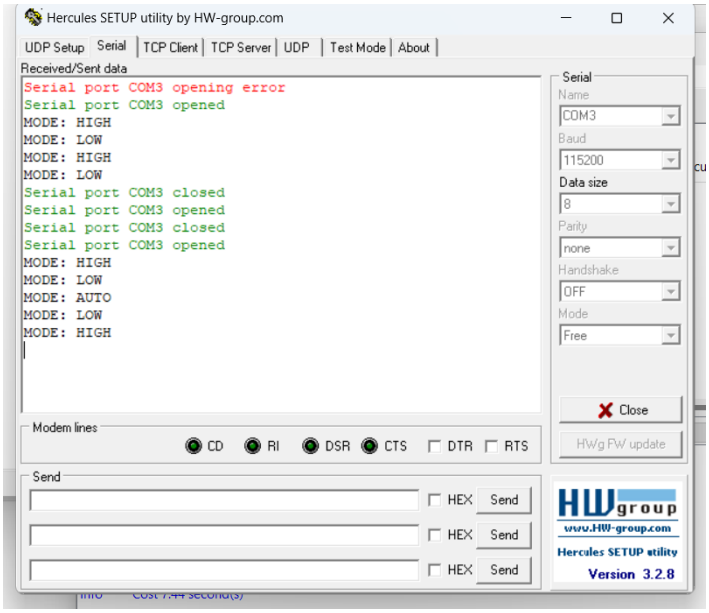


Figure 5.3: UART data received on the Serial Terminal

Result: The host PC successfully receives the status strings via a Serial Terminal such as Hercules or TeraTerm. The sequences ("MODE: LOW\r\n", "MODE: HIGH\r\n", "MODE: AUTO\r\n") are captured perfectly at 115200 bps. No data loss, garbage characters, or framing errors were observed during continuous operation.

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